

- 1.What is design thinking, and why is it important?
- 2.What are the main principles of design thinking?
- 3.Why is empathy a core element of design thinking?

Unit-1

DESIGN THINKING INNOVATION

Design Thinking - Introduction

Design thinking is a methodology that designers use to brainstorm and solve complex problems related to designing and design engineering. It is also beneficial for designers to find innovative, desirable and never-thought-before solutions for customers and clients.

Design thinking is used extensively in the area of healthcare and wellness, agriculture, food security, education, financial services, and environmental sustainability, to name a few. Design thinking has helped in the digital space, contributed to the development of physical products, spurred social innovation projects and much more.

The **iterative design process** helps the designers to involve clients and customers in meaningful ways. It is not just a strategy to come up with feasible solutions to a problem, but also a method to **think of unimaginable solutions** and then trying to **make them not just feasible, but also viable**.

Design thinking is a **blend of logic, powerful imagination, systematic reasoning and intuition** to bring to the table the ideas that promise to solve the problems of the clients with desirable outcomes. It helps to bring creativity with business insights.

Design Thinking - Definition

The idea of using design as a way of solving complex problems in a simplified manner in sciences originated in the book, 'The Sciences of the Artificial', authored by Herbert A. Simon in 1969. The same purpose was achieved for design engineering by the book 'Experiences in Visual Thinking', authored by Robert McKim in 1973.

In 1987, **Peter Rowe's book titled, "Design Thinking"** described methods and approaches that planners, designers, and architects use. The work of Robert McKim was consolidated by Rolf Faste at Stanford University during 1980s to 1990s and then, David M. Kelly adapted design thinking for business interests. David M. Kelly founded IDEO in 1991.

Most of the industries trying to solve customers' problems and address their needs are failing just because they look at the problems outside in. However, many problems can be solved in a better manner if we look at them inside out.

Features of Design Thinking

Such problems require multidimensional solutions. Design thinking helps in this regard. It not only assists a professional to come up with a solution, but it also helps the organization to gain a competitive edge over its rivals. Following are the benefits conferred by design thinking. These are incidentally also the distinguishing features of design thinking.

- Finding simplicity in complexities.
- Having a beautiful and aesthetically appealing product.
- Improving clients' and end user's quality of experience.
- Creating innovative, feasible, and viable solutions to real world problems.
- Addressing the actual requirements of the end users.



Strategy of Innovation

Design thinking is also considered to be a strategy for innovation. It leads to dramatic improvements in innovation. This is why design thinking forms the **core of effective strategy development and seamless organizational change**. Anything that involves human interaction, from products, services, processes etc., can be improved through design thinking. It all depends on the designer's way to create, manage, lead, and innovate.

Use of Design Thinking

The basic principle of design thinking is that innovation can be disciplined. Innovation is not an elusive entity that only a few genius people can experience. It is, rather, a practice that can be systematically approached by a set of practical and meticulous tools, methodologies, and frameworks.

Design thinking helps you learn the following.

- How to optimize the ability to innovate?
- How to develop a variety of concepts, products, services, processes, etc. for endusers?
- How to leverage the diverse ideas of innovation?
- How to convert useful data, individual insights and vague ideas into feasible reality?

- How to connect with the customers and end-users by targeting their actual requirements?
- How to use the different tools used by designers in their profession for solving your customers' problems?

DESIGN THINKING APPLICATIONS

Business

Design thinking helps in businesses by optimizing the process of product creation, marketing, and renewal of contracts. All these processes require a companywide focus on the customer and hence, design thinking helps in these processes immensely. Design thinking helps the design thinkers to develop deep empathy for their customers and to create solutions that match their needs exactly. The solutions are not delivered just for the sake of technology.

Information Technology

The IT industry makes a lot of products that require trials and proof of concepts. The industry needs to empathize with its users and not simply deploy technologies. IT is not only about technology or products, but also its processes. The developers, analysts, consultants, and managers have to brainstorm on possible ideas for solving the problems of the clients. This is where design thinking helps a lot.

Education

The education sector can make the best use of design thinking by taking feedback from students on their requirements, goals and challenges they are facing in the classroom. By working on their feedback, the instructors can come up with solutions to address their challenges.

For example, Michael Schurr, a 2nd grade instructor from New York, realized that his students would be more comfortable with bulletin boards lowered. He also found the idea of creating comfortable semi-private space for working students as it provided them space to study. As a result, his students became more engaged and felt free to move.

Healthcare

Design thinking helps in healthcare as well. The expenditure on healthcare by the government and the cost of healthcare facilities is growing by the day. Experts worldwide are concerned about how to bring quality healthcare to people at low cost.

Venice Family Clinic in Venice, California has come up with innovative solutions to the challenge of opening a low-cost children's clinic to serve the low-income families. Problems of finance, transportation, and language barriers had to be solved. And all this had to be done at low cost for the poor kids. Fostering good health along with profits was a challenge, as it does not sound sustainable. Using design thinking, the inefficiencies in the system and the perennial crises were addressed.

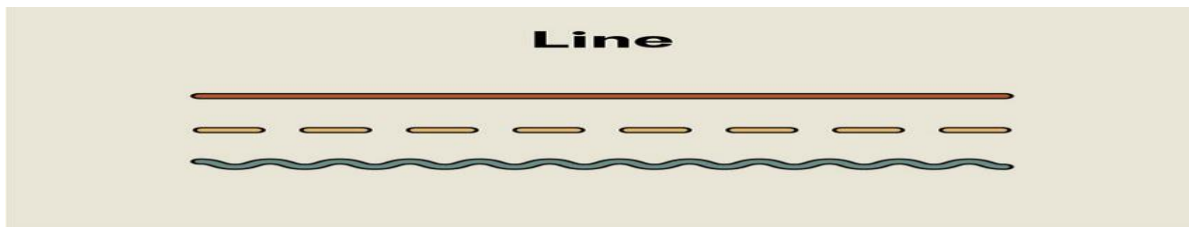
Elements of Design

Think of the elements of design as the ingredients you need for a recipe. These basic components are essential in art and design and how you can visually construct pieces. Each element is a crucial part of a visual message, and the combination of these has an impact on how the design is perceived. You can use these elements alone or in combination with each other, depending on what you're looking to achieve.

The main elements are:

- Line
- Color
- Shape
- Form
- Value
- Space
- Texture

Line

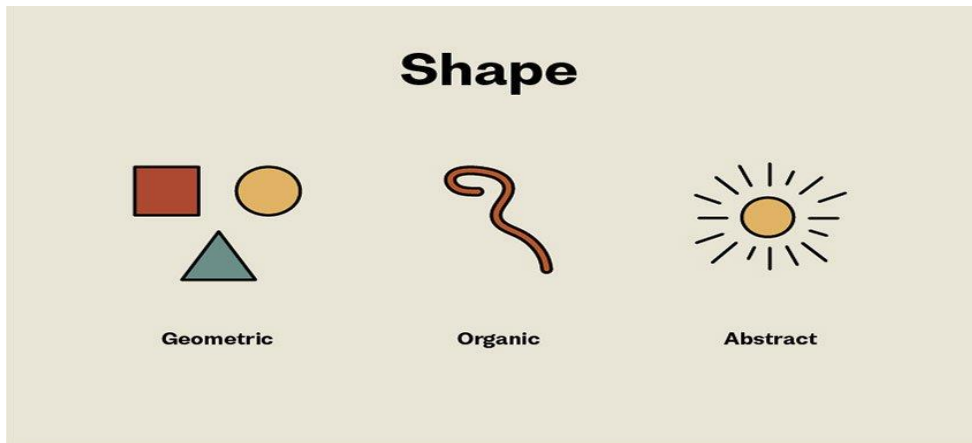


Lines are the most basic elements of design. They come in all shapes, sizes, and colors. Once you start noticing them, you'll see grids all around you. Lines have direction; they can be visible or invisible and can help direct the eye to a specific spot. The thickness of a line can also communicate certain cues. Bold and thick lines can draw attention, while thin lines are the opposite.

Most if not all layouts contain invisible lines. Grids are made of multiple lines and lend structure to a page. Lines can be used to create demarcation on a specific section of a design. Depending on the form of the line, you can convey different moods. A simple line can carry so much—for instance, a squiggly line is perceived as young and fun compared to a straight line.

Lines don't necessarily have to be solid. Dashed and dotted lines can also be used and have a friendlier feel than a solid line. Straight lines usually come across as a steady and static element. On the other hand, curved lines are dynamic and give energy to your design.

Shape



A shape is the result of enclosed lines to form a boundary. Shapes are two-dimensional and can be described as geometric, organic, and abstract.

Geometric shapes have structure and are often mathematical and precise (squares, circles, triangles). You'll notice that the Swiss graphic design movement from the 1950s used mostly geometric shapes in their designs. Shapes can add emphasis to a layout.

Organic shapes lack well-defined edges and often feel natural and smooth. Shapes add emphasis to a layout.

Abstract shapes are a minimalist representation of reality. For instance, a stick figure of a person is an abstract shape. Logos are mostly represented by abstract figures to show the type of business. The icon pack below is a great example of abstract shapes conveying real-life objects and situations.

Form (Positive Space)

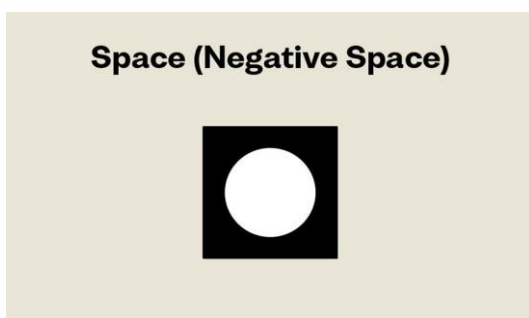


On a page, form is the positive element over the space, the negative element. A dot, line, or shape is a form when placed on a page. Unfortunately, form and shape are

mostly used interchangeably. A form can be either two-dimensional or three-dimensional. Many also believe that form is a shape that acquires three-dimensional values, but the correct term is volume.

Form and shape are mutually dependent because changing one would affect the other. The spatial relationship between form and space can create tension and add 3D qualities to your design. Form and space will lend the design lots of visual activity that can help keep viewers engaged. To create a 3D effect in your design, you can add shadows, stack multiple elements, or play with color.

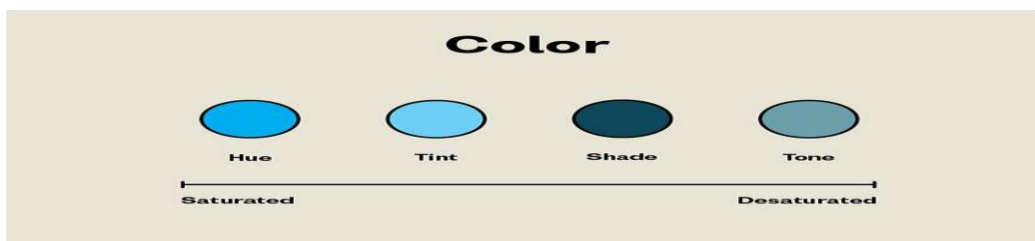
Space (Negative Space)



Space is the area that surrounds a shape; it creates a form within the space. Think of it as music: space is the silence between the notes of a song. If all the notes were played together, that would turn into noise.

If you look at a design piece, the negative space is the area that is not occupied by any elements. In essence, it is the background color that you are able to see. For instance, abundant negative space in a layout results in an open, airy, and light background. The lack of negative space can result in a cluttered design. Visually speaking, a layout needs space to achieve a level of clarity within the design. Negative space is a very important element to consider as you are designing a piece.

Color

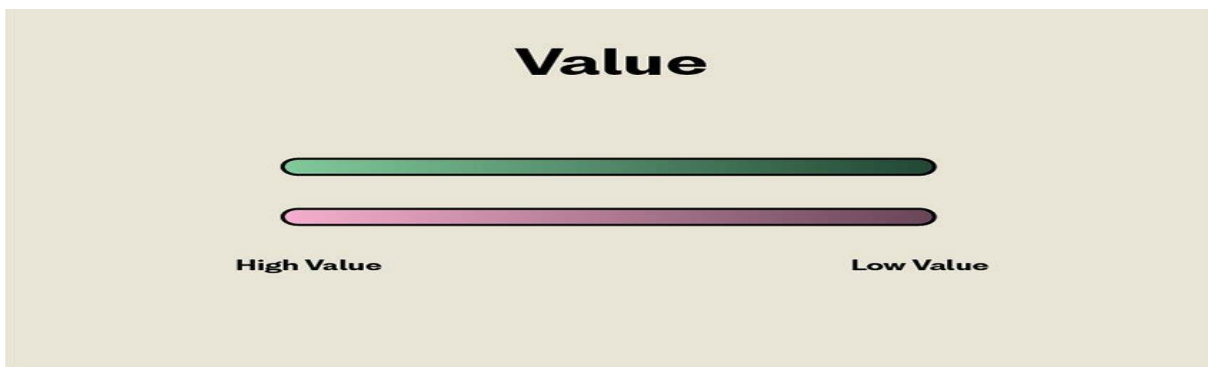


We can apply color to any of the elements we mentioned before this point. Colors create moods and can say something different depending on the connotations associated with it. Color can create an emphasis on specific areas of your design layout.

This element contains multiple characteristics:

- **Hue** is the name of a color in its purest form. For instance, cyan, magenta, and green are pure colors.
 - **Shade** is the addition of black to a hue in order to make a darker version.
 - **Tint** is the addition of white to a color to make a lighter version.
 - **Tone** is the addition of grey to make a color muted.
- **Saturation** refers to the purity of a color. A specific color is most intense when it is not mixed with white or black.

Value



Value refers to the degree of lightness and darkness of a specific hue. Yellow has a higher value than purple because it is closer to white. Value changes create contrast on a page. The reason you can read this text is that the black content contrasts with the white background.

In design, use different tonal values to create emphasis in your design. Create the illusion of movement by overlapping multiple elements with different values. Value is also important in photography. You'll notice that high-value images have a light and airy feel to them, while dark value images feel heavy and dramatic.

Value also defines the spatial relationship between elements. If color values are close between the elements and space, then the design will look flat. If there is a strong contrast between the elements, then the form will be extremely noticeable. The example below features multiple colors with multiple values, which helps add a sense of depth to the design.

Texture

Texture



Texture adds a tactile appearance to a design layout. Imagine how a design piece would feel if you touched it. The goal of texture is to add depth to a 2D surface. Texture can be applied graphically through patterns, either digitally created or an image mimicking the desired pattern. Below is an example of an abstract geometric pattern made up of basic geometric elements.

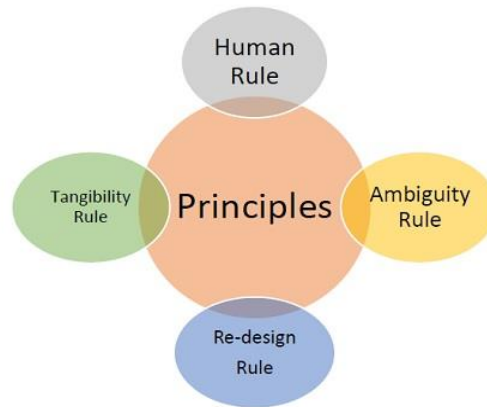
To achieve an authentic vintage feel, you may try a rough effect as a background or a natural pattern like the wood pattern below. Highly texturized patterns like this can translate the feel of wood grains visually. Textures can also be physical—for example, laser cutters give you the ability to stack multiple shapes and intensify a tactile response.

Add real tactile texture to your design by embossing a texture to paper. This way, you have the chance to create a memorable piece that will certainly stand out from the crowd. Stylistically, it is not ideal to blend multiple textures in a design (unless necessary) as it can be overwhelming for the viewer.

Principles of design.

According to Christoph Meinel and Larry Leifer, there are four principles to design thinking.

- **The Human Rule** – This rule states that all kinds of design activity are ultimately social in nature.
- **The Ambiguity Rule** – This rule requires all design thinkers to preserve ambiguity in the process design thinking.
- **The Re-design Rule** – The re-design rule states that all designs are basically examples of re-design.
- **The Tangibility Rule** – The tangibility rule states that making ideas tangible always facilitates communication between design thinkers.



These four principles form the foundation of the design thinking process. A design thinker needs to form his ideas and put them forward based on these principles.

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What are the principles of design thinking?

There are certain principles that are pivotal to design thinking. These are reflected in the design thinking methodology, which we'll explore in detail a little later on. We've outlined five of design thinking's most important principles below.

1. User-centricity and empathy

Design thinking is all about finding solutions that respond to human needs and user feedback. People, not technology, are the drivers of innovation, so an essential part of the process involves stepping into the user's shoes and building genuine empathy for your target audience.

2. Collaboration

The aim of design thinking is to pool a diverse variety of perspectives and ideas; this is what leads to innovation! Design thinking encourages collaboration between heterogeneous, multidisciplinary teams which may not typically work together.

3. Ideation

Design thinking is a solution-based framework, so the focus is on coming up with as many ideas and potential solutions as possible. Ideation is both a core design thinking principle and a step in the design thinking process. The ideation step is a designated judgment-free zone where participants are encouraged to focus on the quantity of ideas, rather than the quality.

4. Experimentation and iteration

It's not just about coming up with ideas; it's about turning them into prototypes, testing them, and making changes based on user feedback. Design thinking is an iterative approach, so be prepared to repeat certain steps in the process as you uncover flaws and shortcomings in the early versions of your proposed solution.

5. A bias towards action

Design thinking is an extremely hands-on approach to problem-solving favoring action over discussion. Instead of hypothesizing about what your users want, design thinking encourages you to get out there and engage with them face-to-face. Rather than talking about potential solutions, you'll turn them into tangible prototypes and test them in real-world contexts.

history of Design Thinking

Drawing on psychological studies of creativity from the 1940s, such as Max Wertheimer's "Productive Thinking" (1945), new creativity techniques in the 1950s and design methods in the 1960s led to the idea of design thinking as a particular approach to creatively solving problems. Among the first authors to write about design thinking were John E. Arnold in "Creative Engineering" (1959) and L. Bruce Archer in "Systematic Method for Designers" (1965), [44][45]

In his book "Creative Engineering" (1959) Arnold distinguishes four areas of design thinking 144 (1) novel functionality, ie, solutions that satisfy a novel need or solutions that satisfy an old need in an entirely new way, (2) higher performance levels of a solution, (3) lower production costs or (4) increased salability (45) Arnold recommended a balanced approach - product developers should seek opportunities in all four areas of design thinking: "It is rather interesting to look over the developmental history of any product or family of products and try to classify the changes into one of the four areas.. Your group, too, might have gotten into

a rut and is inadvertently doing all of your design thinking in one area and is missing good bets in other areas [44]

Although L. Bruce Archer's "Systematic Method for Designers" (1965)⁴⁵ was concerned primarily with a systematic process of designing, it also expressed a need to broaden the scope of conventional design: "Ways have had to be found to incorporate knowledge of ergonomics, cybernetics, marketing and management science into design thinking Archer was also developing the relationship of design thinking with management "The time is rapidly approaching when design decision making and management decision making techniques will have so much in common that the one will become no more than the extension of the other" 147)

Arnold initiated a long history of design thinking at Stanford University, extending through many others such as Robert McKim⁴⁵) and Rotfe Faste ⁴⁵ who taught "design thinking as a method of creative action" 1511 and continuing with the shift from creative engineering to innovation management in the 2000s Design thinking was adapted for business purposes by Faste's Stanford colleague David M. Kelley, who founded the design consultancy IDEO in 1991 153]

Bryan Lawson's 1980 book *How Designers Think*, primarily addressing design in architecture, began a process of generalising the concept of design thinking A 1982 article by Nigel Cross, "Designerly Ways of Knowing established some of the intrinsic qualities and abilities of design thinking that also made it relevant in general education and thus for wider audiences (201 Peter Rowe's 1987 book *Design Thinking*, which described methods and approaches used by architects and urban planners, was a significant early usage of the term in the design research literature. 15 An international series of research symposia in design thinking began at Delft University of Technology in 1991 P Richard Bushanan's 1992 article "Wicked Problems in Design Thinking".